



Research Methods in Psychology

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What is Research? (Group Work-3 mins)

Research



- **Research** is a process of steps used to collect and analyze information to increase our understanding of a topic or issue (Creswell and Creswell, 2018, p:2)
- *Research is a scientific way of knowing the unknown!*

Research (Cont.)



□ Why Research is necessary?

- It corrects human errors and confirmation bias
- It fills gap in knowledge
- It informs evidence-based practice in psychology.



Group Work (3 Mins)

What do these series of fast-food chain indicates about the food preferences and health status of the habitant?

How will you test your assumptions?

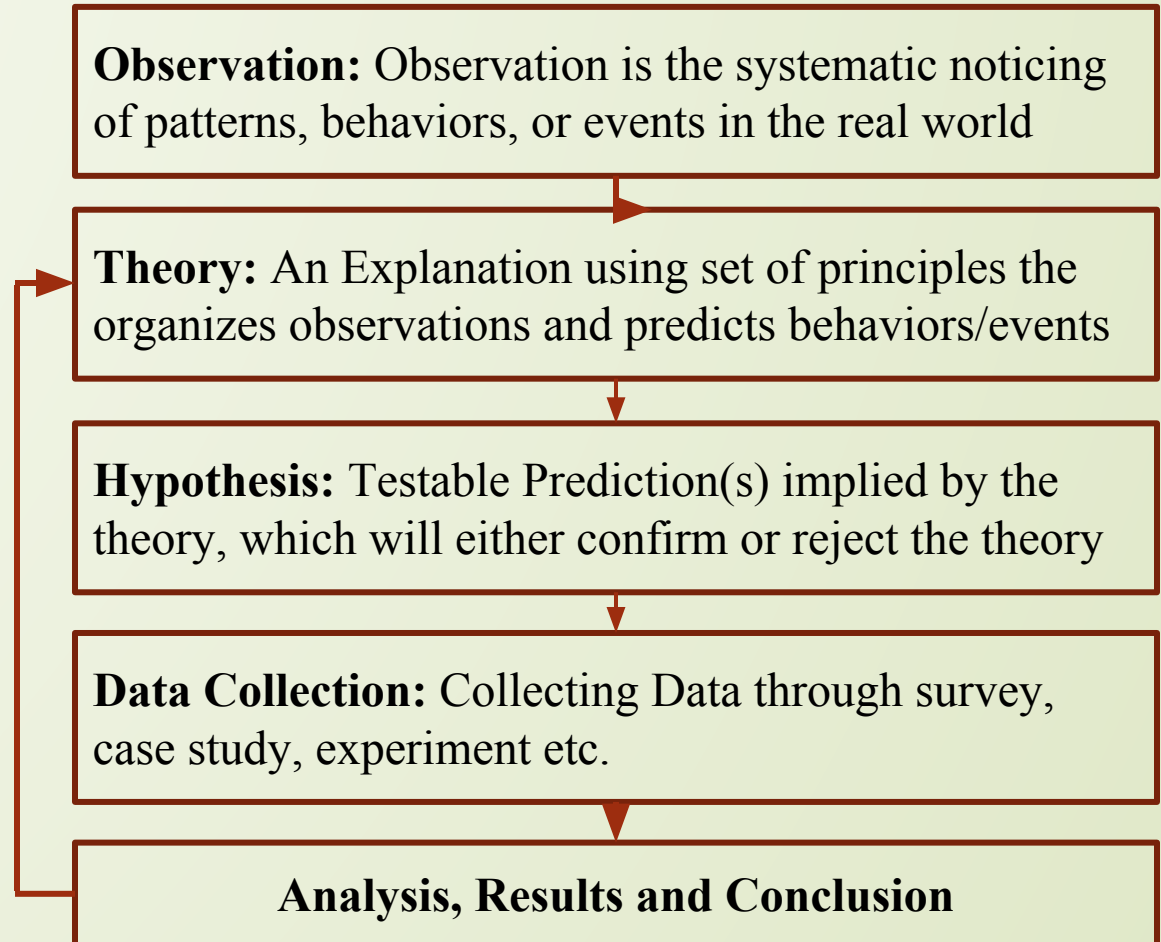
What you just did was critical thinking!



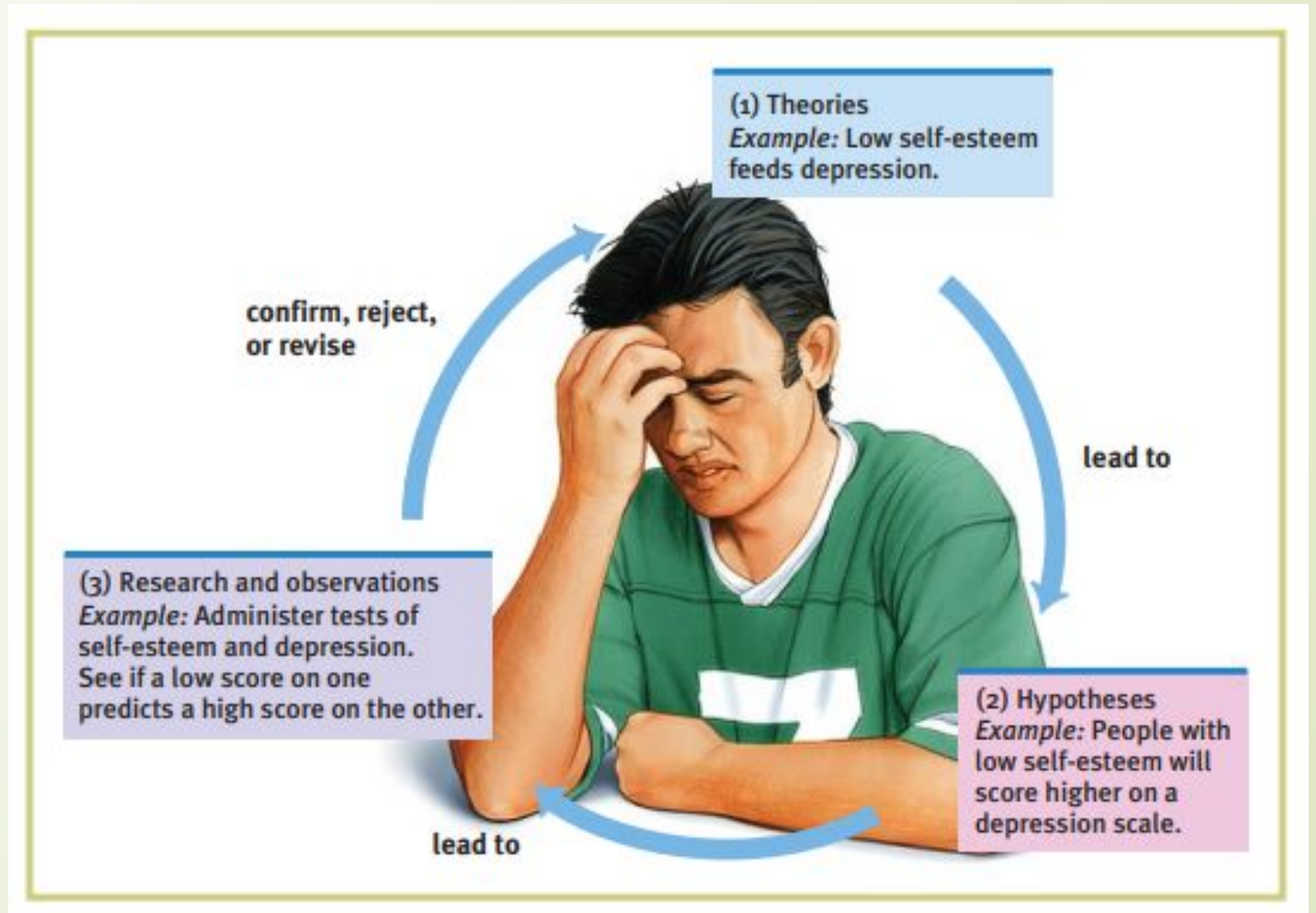
- ❑ **Critical thinkers or Smart Thinkers:**
 - ❑ Do not blindly accept assumptions/arguments and do not jump to conclusions too fast!
 - ❑ Rather, a critical thinker examines assumptions, seeks and evaluates evidence, and assesses conclusions
- ❑ Critical thinking encourages scientific practice and research!

Scientific Methods

- ❑ Scientific method is a self-correcting process for asking questions and observing nature's answer.
- ❑ Psychological researchers make assumptions, test these assumptions against reality, accepting ideas proven to be true, and fixing/rejecting the ones that don't.



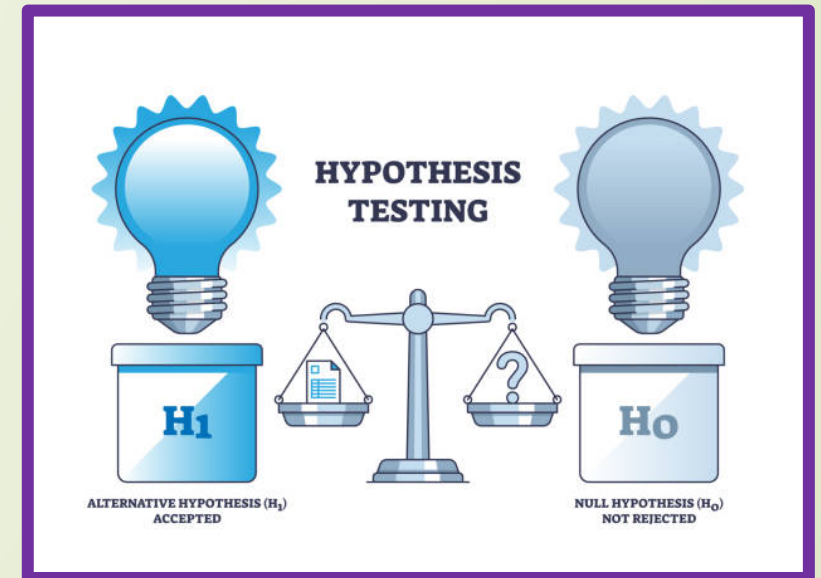
Example



Scientific Method (Cont.)

□ Hypothesis:

- **Null Hypothesis:** States that there is no statistical significance, no difference, no relationship among the variables.
- **Alternative Hypothesis:** There is statistical significance, difference, relationship among the variables.





Null Hypothesis: There is *no difference* between the health status of the habitant of this area and the other neighboring areas.

Alternative Hypothesis: There is *significance difference* between the health status of the habitant of this area and the other neighboring areas



Scientific Method (Cont.)

□ Operational Definition:

- Variables of any research should be operationally defined; there should be ways to measure the variables.
- Example: The health status of the habitants should be measured through body mass index, obesity, blood pressure etc.
- A good research is *replicable, falsifiable* and *reliable*.

Types of Research

1. Experimental
2. Non-Experimental
 - A. Descriptive/Observational Studies (Case study, Survey, Naturalistic Observation)
 - B. Correlational Studies





**Does Counseling Reduces
Depression? (Think-1 min)**

Experimental Design

- Experimental method requires and **control** and can **prove causality**.
- Experimental method measure the effect of independent variable (IV) of dependent variables (DV) to prove whether **IV causes DV**.
- Independent Variable: The experimental factor or variable that has been manipulated, the variable whose affect is being (*Counseling Session would be Independent Variable*)
- Dependent Variable: The outcome variable, variable that might change in response to the manipulation of Independent Variable (*Depression scores would be Dependent Variable*)

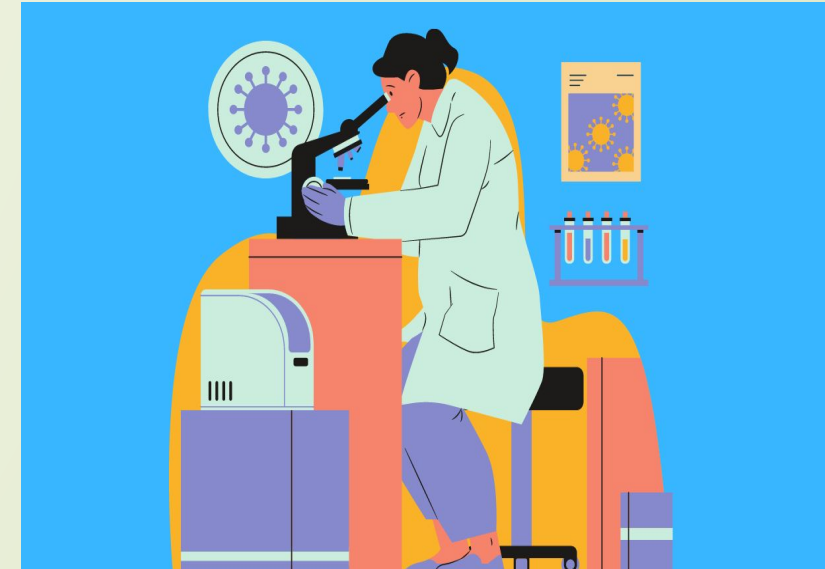
Experimental Design

❑ Experimental Group and Control Group:

- ❑ The group receiving treatment or independent variable is called the experimental group
- ❑ The group which is not receiving treatment or independent variable is called the control group

❑ Random Assignment:

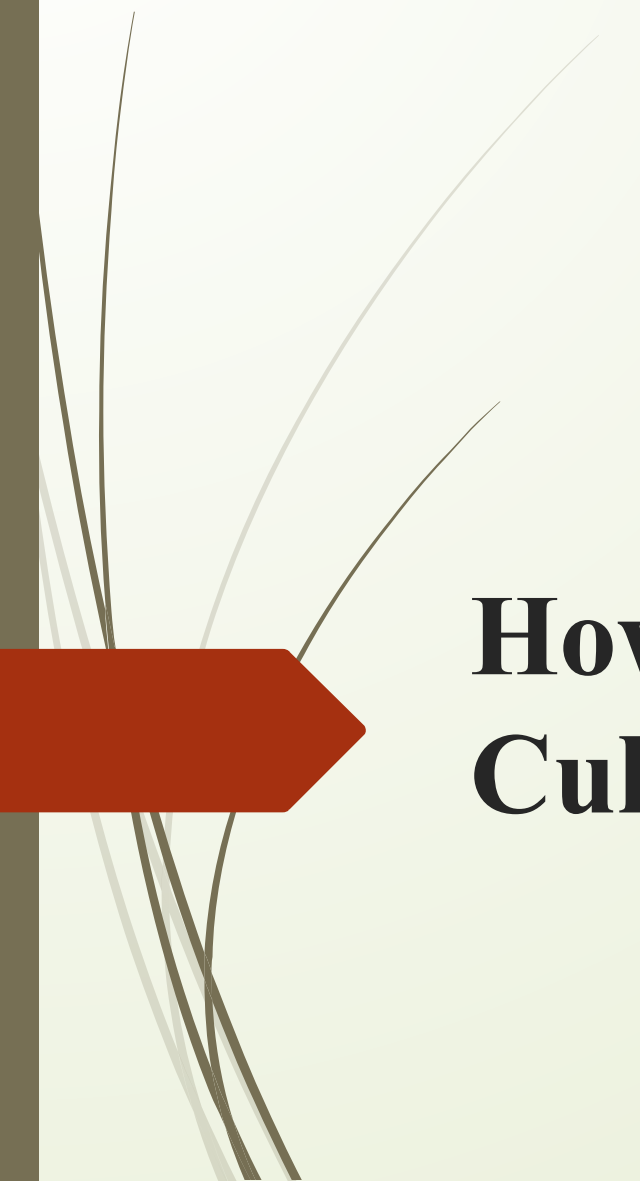
- ❑ Every single participant in your study has an exact equal chance of being placed in the experimental group or the control group. It is entirely decided by chance



Placebo

- ❑ Only the experimental group receives treatment and control group doesn't.
- ❑ Placebo is the any form of treatment that has no significant therapeutic effect of its own (e.g., sugar pills).
- ❑ It is given to the control group to reduce the *expectancy bias*.
- ❑ *Is it ethical to provide Placebo to the control group?*





How Immigrants Adapt in New Culture? (Think-1 min)



Non-Experimental Design: Descriptive

- How can we attempt to answer this question:
 - Naturally observing couple of immigrants in their daily life (e.g., workplace, restaurants, schools, collage, public transport etc.) [**Naturalistic Observation**]
 - Asking as many immigrants about their daily life, coping style etc. [**Survey**]
 - Thorough focusing on handful of people who have been adapting as immigrant in foreign culture, collect detailed account of their stories [**Case Study**]

Case Study

- ❑ A technique in which one person is studied in depth in the hope of revealing universal truth.
- ❑ Earlier studies on human brain and how it functions came from case studies, even today unique/atypical cases are studied deeply to explore new things and ideas!
- ❑ **Caution:** Single cases are not representative of a community! The conclusion coming out of case studies cannot be generalized, it can lead to mislead judgments and false conclusions.



Naturalistic Observation

- ❑ Naturally observing/unobtrusively videotaping people, animals, environment.
- ❑ This technique does not “*explain*” any behavior, it just “*describes*”. However descriptions can also reveal important aspects of life.
- ❑ Cultural and ethnic practices can be best described through observation.
- ❑ **Caution:** Can be very powerful but cannot tell which factors are influencing the behavior/ritual observed.



Survey

- ❑ Survey asks people to report their behavior or opinion, usually by questioning representatives or random sampling of the group.
- ❑ **Caution:** But asking question is tricky, because answers depends on a number of factors:
 - ❑ **Social Desirability Bias:** People give answers that make them look *good*, not necessarily *true*.
 - ❑ **Sampling Bias:** Sampling bias happens when the people you ask do not properly represent the group you want to study.
 - ❑ Not usually preferred for **sensitive/taboo** topics.





Non-experimental Design: Correlational

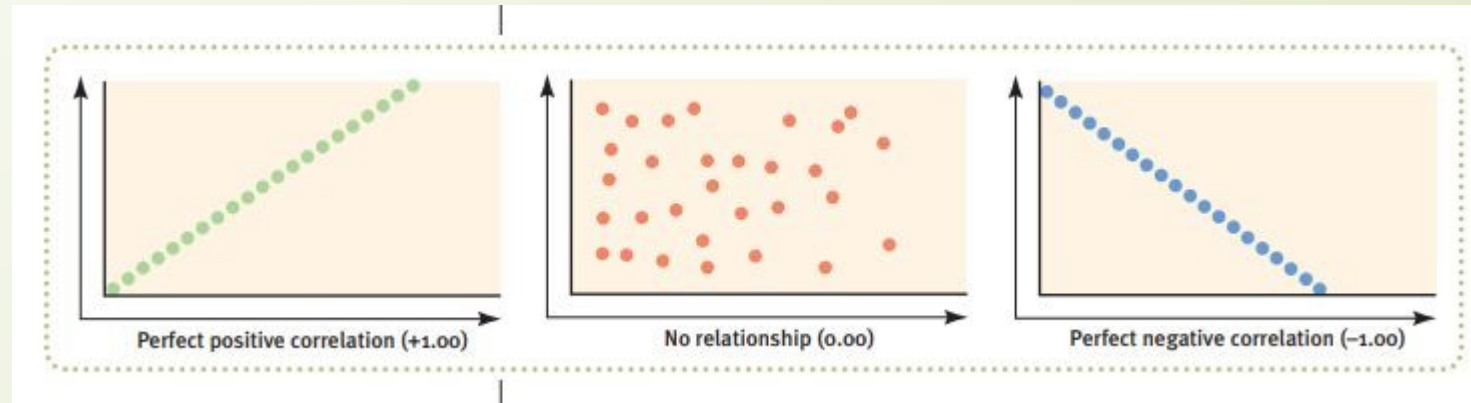
- Correlation is the measure of
 - The extent to which two factors vary together
 - How well either factor predicts the other
- *Can more screen time predict poor academic performance?*
- **Correlation Coefficient:** The statistical index of the relationship among two factors.
 - **Positive Correlation:** Two factors will rise and fall together
 - **Negative Correlation:** One factor going up, other's going down, inversely related

Correlational Research: Is screen-time related to academic performance?

Does Good Pay Enhance Job Satisfaction?

Does frequent aggressive behavior decrease quality social relationship?

Is shoe size related to intelligence?



Correlation and Causation: *No matter how strong the correlation is, it cannot prove causation!*



Types of Research Methodology

Qualitative

- ❑ Studies experiences, meanings, stories, using words NOT NUMBERS!
- ❑ Recruit lesser participants
- ❑ More in-depth
- ❑ Collects data through interview, case studies etc.

Quantitative

- ❑ Studies phenomena using numbers and measurement
- ❑ More participants are recruited for statistically significant results
- ❑ A bit superficial
- ❑ Collects data through survey, experimentation etc.

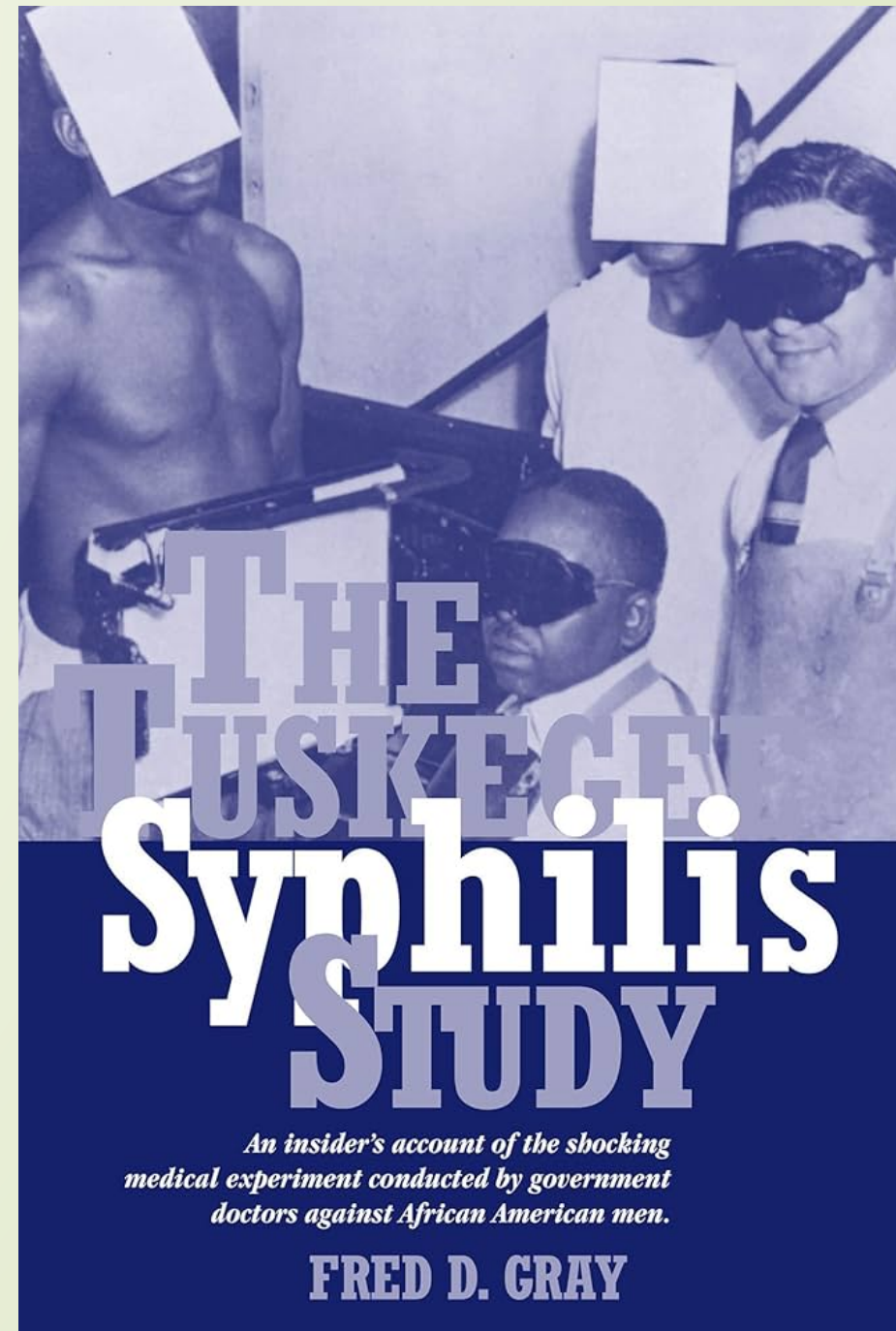


What are the Pros and Cons of Qualitative and Quantitative Methods?

Ethical Consideration in Research

Watch the following video and tell us what is wrong with this research?

[The Tuskegee Trial](#)



Ethical Consideration in Research

- ❑ Informed Consent
- ❑ No False Information and Complete transparency
- ❑ DO NO HARM!
- ❑ Autonomy
- ❑ Confidentiality
- ❑ Voluntary Participation
- ❑ No False Promises!





Research Report Writing

- ❑ **Background and Literature Review** (Explain your variables, and the previous studies on this topic)
- ❑ **Rationale** (Why this research is important?)
- ❑ **Research question** (What question your study is trying to answer?)
- ❑ **Methodology** (Experimental/Non experimental? Qualitative/Quantitative? Justification of your method)
- ❑ **Ethical Considerations**
- ❑ **Results or Findings** (Data, Graphs, Quotations)
- ❑ **Limitations and Implications**



Reflection and Take-Home Message